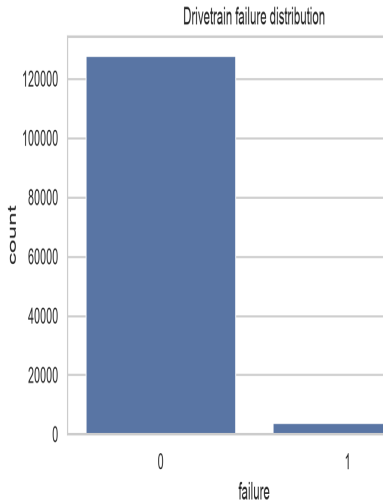
Wind Turbine Failure Detection

Aeolus Renewables

Drivetrain condition classification from 10-minu

Validated timestamp-grouped chronological ev



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Contents / Agenda

1 Executive summary and business problem

2 Exploratory analysis and preprocessing

3 Baseline models, tuning and final selection

4 Final performance, feature interpretation and recommendations

Appendix: data background, baseline table, tuning details and limita

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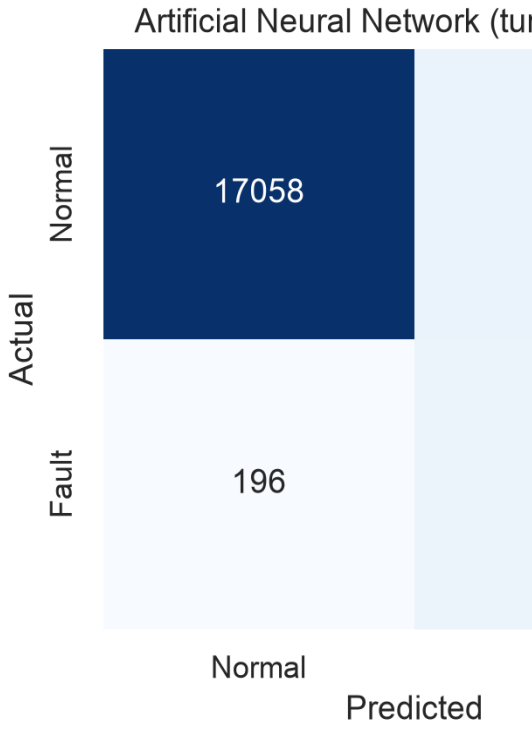
Executive Summary

Tuned ANN selected: 92.49% validation recall

Test recall: 86.40% — 1,245 of 1,441 recorded fault

196 missed states remain the principal risk; 1,271 fa

Deploy as current-condition decision support, not as predictor



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Business Problem Overview and Solution Approach

Business need

Reduce manual SCADA review by prioritising turbines with recorded drivetrain
signatures.

Method

Timestamp-grouped 70/15/15 chronology; training-only imputation/scaling; five
classifiers; tune ANN and XGBoost; select on validation recall; evaluate once o
test.

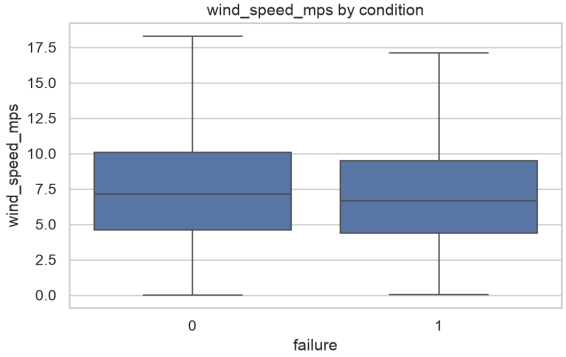
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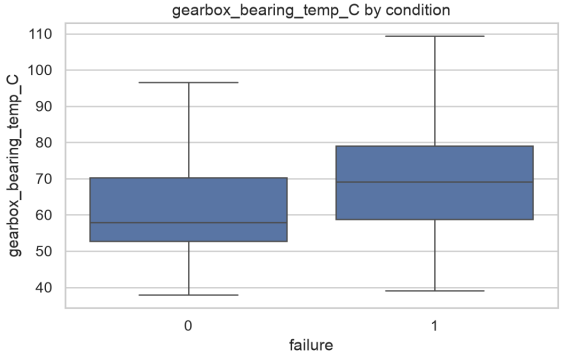
EDA Results — Fault Pattern and Sensor Evidence

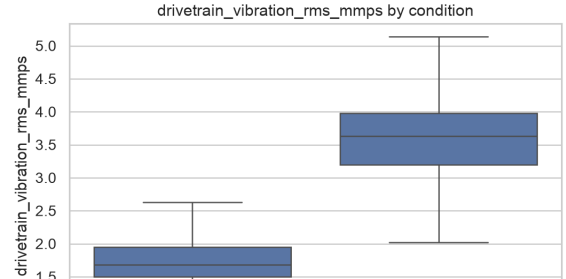
131,760 records from 15 turbines; overall recorded-failure prevalence 3.00%.

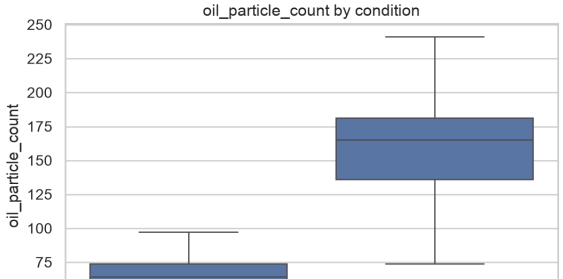
Fault-state distributions shift in gearbox temperature, drivetrain vibration and oil-particle readings.

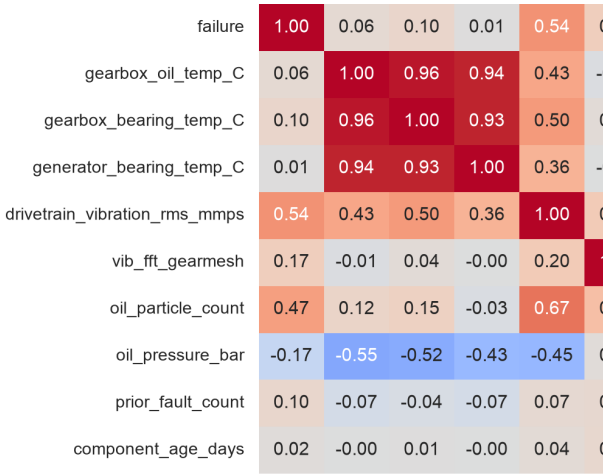
Use signal combinations rather than a single fixed threshold.











tior

Data Preprocessing and Leakage Prevention

Data quality

0 duplicates; 645 gearbox-oil, 690 generator-bearing and 636 oil-pressure missing readings.

Treatment

Median imputation fitted on training rows only; StandardScaler fitted on training row only for ANN.

Features

Calendar cycles, power utilisation, gearbox and generator temperature rise.

Split

Timestamp-grouped chronological partitions prevent the same SCADA timestamp appearing in two splits.

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Model Performance Summary and Final Model Selection

Tuned model	Validation recall	F1	Precision
ANN	92.49%	53.54%	37.67%
XGBoost	69.10%	69.34%	69.57%

Selection rule: highest validation recall; F1 then precision only if recall ties.

ANN wins on recall; XGBoost is more precise and has better F1, but misses more recorded faults.

Test data did not influence tuning, selection or threshold choice.

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Actionable Business Recommendations

1. Rank persistent high-risk alerts for operations-centre review.
2. Check vibration, thermal, oil and maintenance context before dispatch.
3. Send recurring or multi-signal alerts to planned inspection queues.
4. Monitor missed states, false-alert workload, sensor drift and downtime outcomes.
5. Validate prospectively before claiming lead time or financial savings.

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APPENDIX — Data Background and Contents

SCADA panel data: 8,784 ten-minute timestamps across 15 turbines.

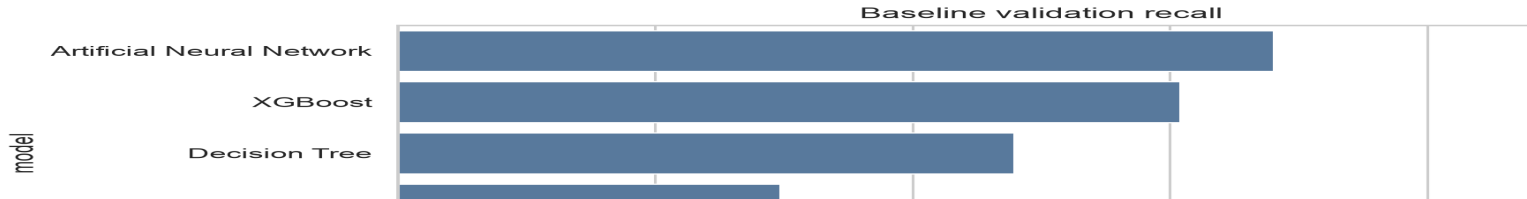
Input groups: weather, production, rotational speed, temperatures, vibratio
spectra, oil condition, operating history and maintenance context.

Target: failure indicates the condition recorded at the timestamp.

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Model Building — Baseline Validation Comparison

Model	Accuracy	Precision	Recall
Decision Tree	90.7%	27.7%	47.9%
Random Forest	95.3%	82.9%	14.4%
Gradient Boosting	94.5%	47.0%	29.7%
XGBoost	96.6%	71.0%	60.8%
Artificial Neural Network	94.8%	50.3%	68.0%



Model Improvement — Hyperparameter Tuning

XGBoost search: 6 randomized configurations; two timestamp-grouped expanding f
recall optimization.

Ranges: 80–200 trees, depth 3–6, learning rate 0.02–0.10, row/column sampling 0.
1.00, min-child weight 1–5.

Best XGBoost CV recall 48.45%; holdout validation recall 69.10%.

ANN: 64/24 layers, BatchNorm, dropout 0.30/0.15, Adam and early stopping.

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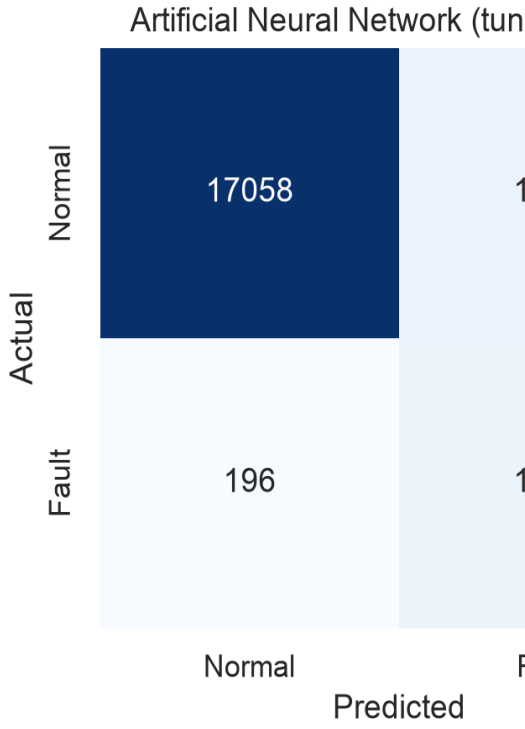
Final Model Performance — Untouched Test Set

Tuned ANN

Accuracy 92.58% | Precision 49.48% | Recall 86.40% | F

TN 17,058 | FP 1,271 | FN 196 | TP 1,245

Interpretation: false negatives require monitoring; false po
and alert-persistence rules.



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Feature Interpretation and Deployment Limitations

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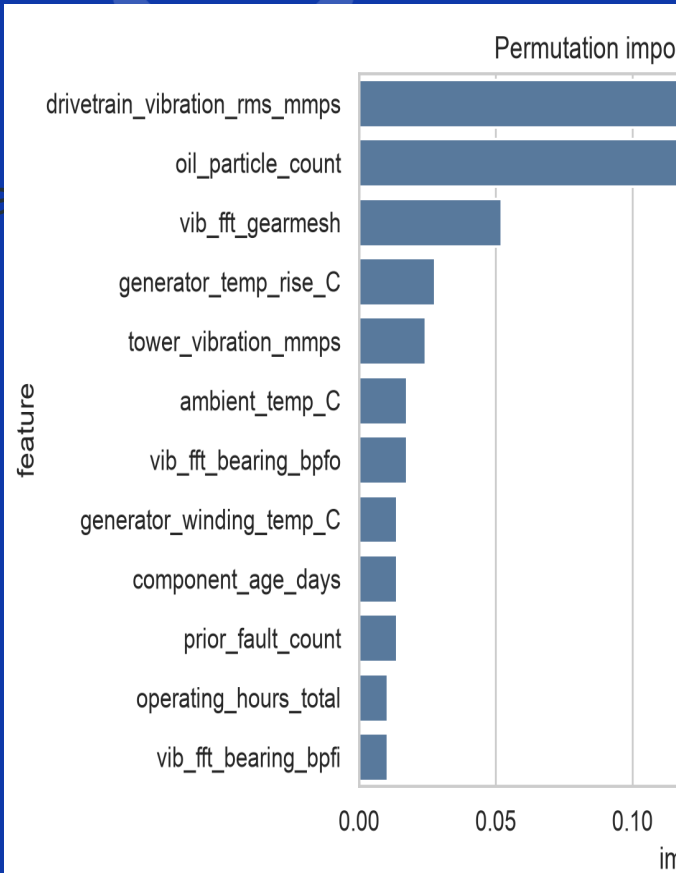
Permutation importance measures validation-recall

It identifies associated condition signals, not causal

Use top signals to focus technician checks; retrain a

change.

Happy Learning !



Timestamp-Grouped Split Verification

Train: 92,220 rows | 6,148 timestamps | 1.60% faults

Validation: 19,770 rows | 1,318 timestamps | 5.26% faults

Test: 19,770 rows | 1,318 timestamps | 7.29% faults

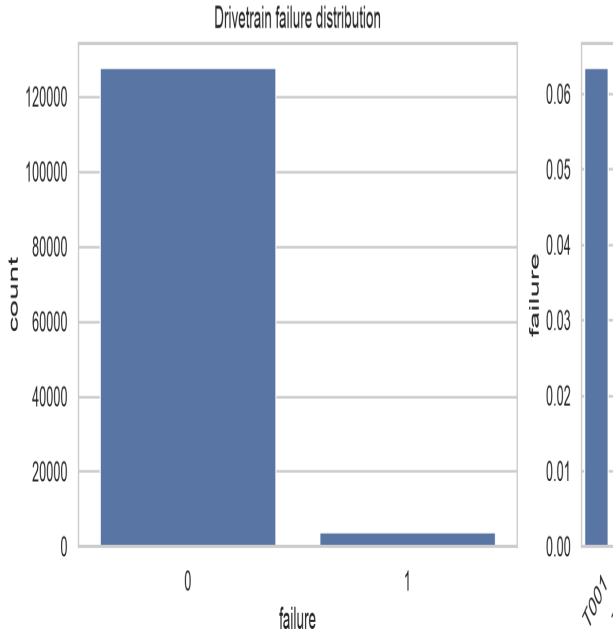
No timestamp occurs in more than one partition.

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EDA Supporting View

Class distribution and turbine-level failure patterr
classification task.

Failure prevalence differs over time; accuracy alc



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Operational Use Case

Operations centre reviews alert evidence remotely.

Technician dispatch follows persistence or corroborating signals.

Feedback from work orders supports recalibration and governance.

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